LAB WORKSHEET

Fault-Finding Lab

Find and fix planted faults in a network using a systematic, evidence-based method.

Method (bottom-up)

- **Layer 1:** link lights, `show ip interface brief`, `show controllers serial 0/0/0`, Physical view.
- **Layer 2:** `show vlan brief`, `show interfaces trunk`, `show cdp neighbors`.
- **Layer 3:** `ipconfig /all`, `ping`, `tracert`, `show ip route`, `show ip protocols`.
- **Layers 4–7:** `show access-lists`, `nslookup`, `ipconfig /renew`.

Typical fixes

```
no shutdown
switchport access vlan 10
switchport mode trunk / switchport trunk native vlan 99
clock rate 128000 / encapsulation hdlc
ip route 0.0.0.0 0.0.0.0 203.0.113.1
network 192.168.20.0 0.0.0.255 area 0
ip helper-address 192.168.50.10
```

Evidence

One log row per fault: **symptom → test → cause (OSI layer) → fix → re-test**. Then Copy report.